

Experiment - 8

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Section: KRG-2B

Problem Statement

Given two strings `str1` and `str2`, return the shortest string that has both `str1` and `str2` as subsequences. If multiple answers exist, return any one of them.

Input

`str1 = "abac"`

`str2 = "cab"`

Output

`cabac`

Approach

Use Dynamic Programming.

First compute the Longest Common Subsequence (LCS) of the two strings.

Create a DP table where `dp[i][j]` stores LCS length of first `i` and `j` characters.

Traverse the table from bottom-right to construct the answer.

If characters match → include once.

If not → take character from the direction with larger LCS.

Add remaining characters if any.

Reverse the final string to get the result.

Pseudo Code

1. Let `n = length of str1`, `m = length of str2`

2. Create `dp[n+1][m+1]` initialized with 0

3. Fill dp table:

For `i` from 1 to `n`:

For `j` from 1 to `m`:

If $str1[i-1] == str2[j-1]$:

$dp[i][j] = 1 + dp[i-1][j-1]$

Else:

$dp[i][j] = \max(dp[i-1][j], dp[i][j-1])$

4. Build result string:

$i = n, j = m$

$ans = ""$

While $i > 0$ and $j > 0$:

If characters match:

add character to ans

$i--, j--$

Else if $dp[i-1][j] > dp[i][j-1]$:

add $str1[i-1]$, $i--$

Else:

add $str2[j-1]$, $j--$

5. Add remaining characters of $str1$ or $str2$

6. Reverse ans

7. Return ans

Time Complexity

$O(n \times m)$

Space Complexity

$O(n \times m)$